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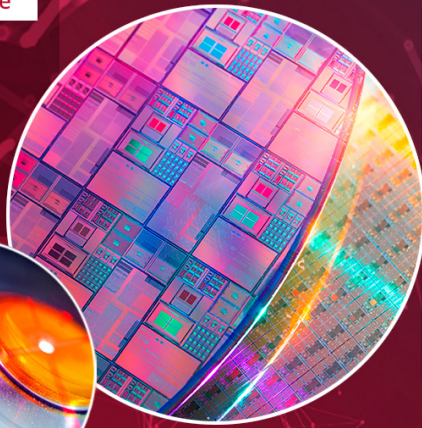


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Gas rarefaction and the time evolution of long high-power impulse magnetron sputtering pulses

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Abstract

Model studies of 400 μs long discharge pulses in high-power impulse magnetron sputtering have been made to study the gas dynamics and plasma chemistry in this type of pulsed processing plasma. Data are taken from an experiment using square voltage pulses applied to an Al target in an Ar atmosphere at 1.8 Pa. The study is limited to low power densities, $<0.5 \text{ kW cm}^{-2}$, in which the discharge is far away from the runaway self-sputtering mode. The model used is the ionization region model, a time-dependent plasma chemistry discharge model developed for the ionization region in magnetron sputtering discharges. It gives a close fit to the discharge current during the whole pulse, both an initial high-current transient and a later plateau value of constant lower current. The discharge current peak is found to precede a maximum in gas rarefaction of the order of $\Delta n_{\text{Ar}}/n_{\text{Ar},0} \approx 50\%$. The time durations of the high-current transient, and of the rarefaction maximum, are determined by the time it takes to establish a steady-state diffusional refill of process gas from the surrounding volume. The dominating mechanism for gas rarefaction is ionization losses, with only about 30% due to the sputter wind kick-out process. During the high-current transient, the degree of sputtered metal ionization reaches 65–75%, and then drops to 30–35% in the plateau phase. The degree of self-sputtering (defined here as the metal ion fraction of the total ion current to the target) also varies during the pulse. It grows from zero at pulse start to a maximum of 65–70% coinciding in time with the maximum gas rarefaction, and then stabilizes in the range 40–45% during the plateau phase. The loss in deposition rate that can be attributed to the back-attraction of the ionized sputtered species is also estimated from the model. It is low during the initial 10–20 μs , peaks around 60% during the high-current transient, and finally stabilizes around 30% during the plateau phase.

(Some figures may appear in colour only in the online journal)

1. Introduction

High-power impulse magnetron sputtering (HiPIMS) is an emerging technology using high-power unipolar pulses of

short duty cycle applied to the cathode in a conventional magnetron deposition system [1]. It was introduced by Kouznetsov *et al* [2] in 1999. Typical HiPIMS operational parameters are peak target power density $P_{\text{target}} > 0.5 \text{ kW cm}^{-2}$, duty cycle

0.5–5%, pulse repetition frequency 50 Hz–5 kHz, pulse length 5–500 μ s, target voltage up to 2000 V and peak current densities up to 3–4 A cm⁻² [1, 3]. Due to the low duty cycle the average power is relatively low, and the target heating can be kept within acceptable limits, even when the discharge has a peak plasma density above 10¹⁹ m⁻³ and a high ionization fraction of the sputtered species [4, 5]. HiPIMS is an emerging tool for self-ion-assisted deposition of thin films, nanocomposites, hard coatings, functional coatings for electronic applications or coatings on complex surfaces [1, 2, 6–9]. The method shows promising results regarding tailoring of the microstructure as well as improved thin film properties, such as improved adhesion [6] and increased wear resistance [10–12].

With pulsed operation the parameter space is considerably widened compared with conventional magnetron sputtering, since the peak power, the pulse length and the pulse repetition frequency can be independently varied. In addition to these, the shape of the pulse can be tailored and modulated to give a pulse train sequencing. The only constraint is that the average power density to the target must be below the damage limit, typically 0.05 kW cm⁻² but depending on the target cooling efficiency. The optimization of the pulse characteristics is therefore a difficult and time-consuming task, particularly so because the desired properties can vary with the application in question. Here we focus on four parameters relevant for such an optimization, and how these vary during the pulse: the degree of process gas rarefaction $\Delta n_{Ar}/n_{Ar,0}$, the degree of ionization $n_{M^+}/(n_M + n_{M^+})$ of the sputtered metal species M , the degree of self-sputtering, and the reduction in the deposition rate due to electric back-attraction of M^+ ions.

In this work we combine experiments and modeling. The experiments were performed at the Lawrence Berkeley National Laboratory, using a square voltage HiPIMS pulse, 400 μ s long, applied to an Al target in an Ar atmosphere at 1.8 Pa [13]. We limit the study to current densities at the cathode below 1.25 A cm⁻², and power densities so low that the discharge is well below the self-sustained self-sputtering mode [13]. The model used is the ionization region model (IRM) [14], a time-dependent plasma chemistry discharge model developed for the ionization region that hovers next to the cathode target of magnetron sputtering discharges. Here this model is modified and adapted to the present discharge geometry. This paper is organized as follows. The experiment is described in section 2 and the fitting of the IRM to the experimental data in section 3. Section 4 contains an analysis of the time development during the discharge pulse. Gas rarefaction is treated in section 4.1, the plasma parameters n_e and T_e , and the ionization fraction of the sputtered species in section 4.2, the degree of self-sputtering in section 4.3 and a discussion of the deposition rate reduction in section 4.4. Section 5 contains a discussion and a summary.

2. Experiment

The experiments reported by Anders *et al* [13] and used here as input to the model work were made using a planar magnetron, equipped with a 50 mm diameter, 6.25 mm thick circular Al target. The magnetic field strength was 64 mT at the target

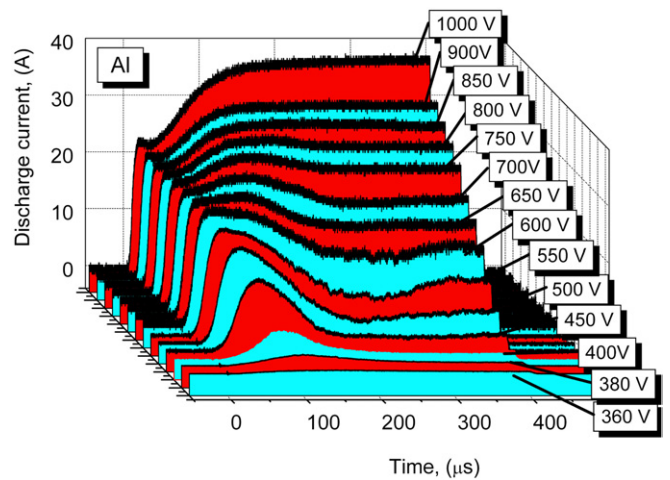


Figure 1. Experimentally determined temporal variation of the discharge current for various cathode voltages. The cathode target is Al, 50 mm in diameter (from [13]).

surface center, and lower elsewhere. The circular race track diameter was 25 mm. The vacuum chamber was pumped with a cryogenic pump to a background pressure of about 10⁻⁴ Pa. Pure argon was supplied near the target, establishing an operational pressure of 1.8 Pa, which was monitored by an MKS Baratron gauge. High-voltage pulses were applied by a SPIK2000A pulse power supply (Melec GmbH) operating in the unipolar negative mode at constant voltage, capable of delivering up to 1000 V and 120 A peak values with pulse duration lengths of up to several milliseconds. Further details on the experimental conditions can be found elsewhere [13].

The temporal variation of the discharge current $I_D(t)$ for various constant voltages $U_D(t)$ for discharges with the Al target is shown in figure 1. During the pulses, the system undergoes a transition from the initial pressure-dependent and gas-dominated phase to a later power- and material-dependent phase that includes self-sputtering. A transition to a high level of current starts at about 550 V, and at voltages higher than around 700 V the current at later time gradually exceeds the initial peak until it reaches a higher and relatively stable value. We focus in this work on discharges far below this high-current range, and therefore concentrate on pulses with $U_D \leq 450$ V for the modeling task.

3. Fitting of the model to experimental data

The model used in this work, the IRM [14, 15], is based on a separation of the discharge into different regions, as shown in figure 2. It is a zero-dimensional (global) plasma chemistry model of the ionization region and of the particle fluxes across its boundaries. The time development is defined by a set of ordinary differential equations giving the first time derivatives of the electron temperature T_e , and of the particle densities of the sputtered target species (metal) atoms n_M as well as the singly charged ions n_{M^+} , and the process gas (argon) ions n_{Ar^+} . The total argon gas density n_{Ar} is subdivided into cold neutrals $n_{Ar,c}$, metastable $n_{Ar,m}$, hot $n_{Ar,h}$ and warm $n_{Ar,w}$. The latter two consist of recombined argon ions that come back from the

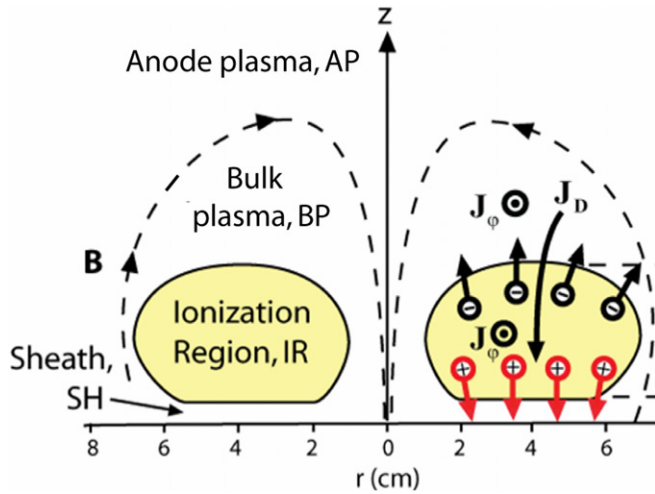


Figure 2. Separation of the discharge into regions including sheath (SH), ionization region (IR) and bulk plasma (BP), from [14]. The ionization region is defined as the volume above the race track where most of the ionization occurs.

target as neutrals and are split up into a hot population n_{Ar^H} that immediately returns to the discharge volume, and a warm population n_{Ar^w} that first penetrates into the target, and later diffuses to the surface and leaves with the target temperature $T_t \approx 0.1$ eV. The electron density n_e is determined from the sum of the ion densities by assuming quasi-neutrality of the plasma. As described by Raadu *et al* [14], the model is adjusted to a particular device by defining an annular IRM volume that has a radial width $r_1 - r_2$ of the race track and an axial extent $z_1 - z_2$ that is estimated from the magnetic topology and/or optical and probe data on the plasma profile. This geometry comes into the model in the form of particle fluxes that are proportional to the boundary areas in question, $\Gamma_{NN}S_{RT}$ to the race track, and $\Gamma_{NN}S_{BP}$ to the bulk plasma (where the index NN denotes the species). The net loss (or gain) from these fluxes is averaged over the volume V_{IR} to fit in a global description. One also needs the reaction set for the working gas and target species. For an aluminum target in argon gas we use the reaction set tabulated in [14]. As geometric approximation we take the race track to be 12.7 mm wide (inner radius 6.1 mm and outer radius 18.8 mm), and extending outwards from the target surface as defined by $z_1 < z < z_2$, where the sheath thickness is $z_1 = 0.2$ mm, and the extent from the target surface is $z_2 = 6.35$ mm. We opt for modeling a square voltage, -450 V pulse, and the corresponding experimentally determined discharge current $I_D(t)$ shown in figure 1. They are fitted by analytical functions. The discharge voltage $U_D(t)$ is used as an input variable and therefore an energetically self-consistent model run is obtained when modeled and experimentally determined discharge currents agree, i.e. $I_{calc}(t) = I_D(t)$. The model current is calculated at the target surface as [14]

$$I_{calc}(t) = e \cdot S_{RT} \Gamma_{Ar^+} (1 + \gamma_{Ar^+}) + e \cdot S_{RT} \Gamma_{M^+} (1 + \gamma_{M^+}). \quad (1)$$

Here, Γ_{Ar^+} and Γ_{M^+} are the ion flux densities as they enter the cathode sheath with the ion acoustic speed, S_{RT} is the area of the race track, and γ_{Ar^+} and γ_{M^+} are the secondary electron emission probabilities. The energy of recombined

argon ions that are reflected has previously been studied in the case of direct current magnetron sputtering (dcMS). Vink and van Zon [16] report energies in the range of several tens of eV when sputtering Mo (95.9 amu) using Ar, while Yamamura and Ishida [17] give lower values, approximately 5–10 eV, for the pressure used in this study, when modeling Cu (63.6 amu) sputtering in Ar using a Monte Carlo simulation. Both studies conclude that the precise value is sensitive to operating pressure and discharge voltage. The energies are likely to be even lower in Al (27.0 amu) as used here, due to the lower target atomic mass. We adapt the value 2 eV used in [14], which was based on computer simulations made specifically by K R Lee for the purpose. In the first runs of the model we assume all recombined argon atoms to be directly reflected, i.e. $n_{Ar^w} = 0$. We will show below in section 4.1 that this is not a crucial assumption.

As described by Brenning *et al* [15], two model fitting parameters are used. The first fitting parameter, F_{PWR} , is the fraction of the total discharge power $P_D(t) = I_D(t) \cdot U_D(t)$ that goes to energize the electrons in the ionization region, $F_{PWR} = P_e(t)/P_D(t)$. The second fitting parameter is the probability β for ions to leave the ionization region toward the target; the same value of β is used here for Ar^+ and M^+ ions. This gives a two-dimensional parameter space for model fitting. We first estimate these parameters from the potential distribution in the device.

An emissive probe has recently been developed by Sanders *et al* [18] and Rauch *et al* [19] for measurements of plasma potential in dense HiPIMS plasmas. Preliminary measurements in our device (not shown here) reveal a potential drop across the ionization region $U_{IR} \approx 50$ V and an additional few tens of volts extending some centimeters further into the bulk plasma. (This is a general phenomenon; similar potential profiles have also been reported previously by Mishra and Bradley [20].) The value $U_{IR} \approx 50$ V can be used for preliminary estimates of both β and F_{PWR} as follows. Regarding the β value, it is on the one hand well known that both argon and metal ions do escape from the ionization region [21]. Both types of ions fill the bulk plasma, and are also observed at the substrate and the walls [22]. Consequently, β must be at least some fraction below 1. On the other hand, the potential profile described above gives a potential barrier, uphill in the z direction away from the target, which from most points of ionization is larger than the typical kinetic energy of the ions. Most ions should therefore be attracted back to the target. We take $\beta > 0.8$ as a lower limit, giving a likely range of $0.8 < \beta < 1$. Regarding the F_{PWR} value, the part of the total discharge power $I_D U_D$ that goes to energize the electrons is given by the integral $\int \mathbf{J}_e \cdot \mathbf{E} dV$ over the whole discharge volume V , where $\mathbf{J}_e$ is the electron part of the current density. We assume that most electron energization occurs in the sheath and the ionization region, and introduce the average electron currents in these regions $I_{e,SH}$ and $I_{e,IR}$. This gives the power fraction to the electrons as

$$F_{PWR} = \frac{P_e}{P_D} = \frac{I_{e,SH} \cdot U_{SH} + I_{e,IR} \cdot U_{IR}}{I_D \cdot U_D}. \quad (2)$$

In dcMS [23], there can be a significant amount of ionization in the sheath. However, in the denser HiPIMS plasma, the

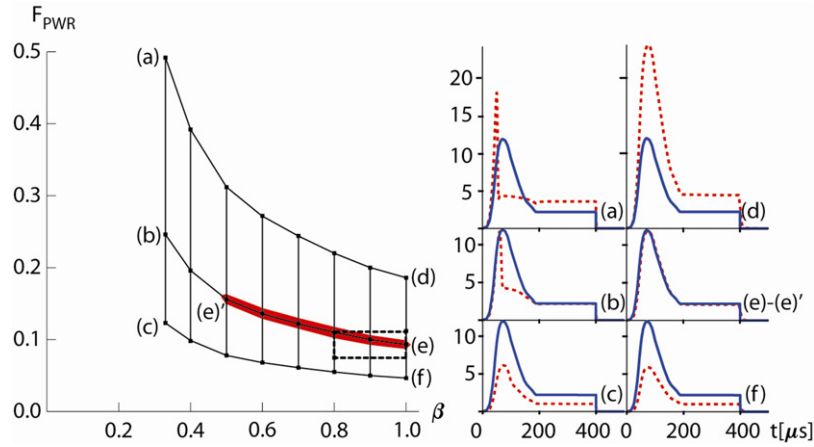


Figure 3. Left panel: the line (b)–(e) shows, for given values of the ion back-attraction probability β , the value of the electron energy fraction F_{PWR} where the IRM gives the best overall fit of the experimental discharge current $I_D(t)$. The red thick line (e)–(e)' shows the range with good fits of the whole pulse; one such good fit is exemplified by the right panel marked (e)–(e)'. The dashed rectangle defines the independently estimated range of β and F_{PWR} from the experimentally determined potential profile. Right panel: curves of the experimental discharge current (blue), and the modeled (red dashed) calculated for the combinations of fitting parameters (β , F_{PWR}) marked (a)–(f) in the left panel.

sheath is much thinner. We therefore neglect ionization in the sheath and approximate $I_{e,\text{SH}}$ to be carried by secondary emission electrons only. For Ar^+ ions sputtering aluminum, with $U_{\text{SH}} \approx 400$ V, the secondary emission coefficient is $\gamma_{\text{SE}} \approx 0.09$ [24]. The emitted electrons are, however, trapped on the curved magnetic field lines, making some of them return and become recaptured at the target surface. This is taken into account by a recapture probability r , and an effective coefficient of secondary electron emission $\gamma_{\text{EFF}} = (1 - r) \cdot \gamma_{\text{SE}}$. The recapture probability r is often assumed to be around 0.5 [25, 26]. However, Buyle *et al* [27] and Costin *et al* [28] showed that there is a significant difference between individual electrons depending mainly on the distance, from where they are emitted, to the race track center. Based on their results, we use an uncertainty range of the r value from 0.25 to 0.75. The electron current in the sheath is thus $I_{e,\text{SH}} \approx I_1 \cdot \gamma_{\text{EFF}} = I_D \cdot \gamma_{\text{EFF}} / (1 + \gamma_{\text{EFF}})$, where I_1 is the ion current to the target and $I_D = I_1 \cdot (1 + \gamma_{\text{EFF}})$. To estimate $I_{e,\text{IR}}$ in the ionization region, we note that the electron fraction of the current depends on the position, as illustrated in figure 2. It increases gradually from $I_D \cdot \gamma_{\text{EFF}} / (1 + \gamma_{\text{EFF}})$ at the sheath edge to approximately the full I_D at the boundary to the bulk plasma. We take 0.5 for an average, i.e. $I_{e,\text{IR}} \approx \frac{1}{2} I_D$. Based on the above discussion and equation (2), we get

$$F_{\text{PWR}} \approx \frac{I_D \cdot (\gamma_{\text{EFF}} / (1 + \gamma_{\text{EFF}})) \cdot U_{\text{SH}} + \frac{1}{2} I_D \cdot U_{\text{IR}}}{I_D \cdot U_D} = \frac{(\gamma_{\text{EFF}} / (1 + \gamma_{\text{EFF}})) \cdot U_{\text{SH}} + \frac{1}{2} U_{\text{IR}}}{U_D}. \quad (3)$$

With $U_{\text{SH}} \approx 400$ V, $U_{\text{IR}} \approx 50$ V, $U_D \approx 450$ V, $\gamma_{\text{SE}} \approx 0.09$, and using the limits $0.25 < r < 0.75$, we obtain the range $0.075 < F_{\text{PWR}} < 0.112$. The resulting combined ranges of F_{PWR} and β are marked with a dashed rectangle in the left-hand panel of figure 3.

A completely independent estimate is obtained using the model as follows. For each pair (β , F_{PWR}) of the fitting parameters, the model gives a calculated current waveform

$I_{\text{calc}}(t)$. Some examples are shown in the right-hand panel in figure 3. Only in the range $0.5 < \beta < 1$ and on the curve marked (e)–(e)' could good fits be obtained of both the high-current peak (shape, amplitude and time duration) and of the value of the current plateau. This, therefore, represents the range of possible discharge conditions according to the IRM, i.e. those for which the development of the discharge current during the pulse is self-consistent with regard to the requirements of energy conservation, the reaction data set and all the separate particle balances.

To conclude, from the experimentally determined potential profile we estimate the (β , F_{PWR}) ranges marked by a dashed rectangle in figure 3. They agree with, and represent a subset of the good fitting range of, the IRM. In the following sections we will stay in this part of the (β , F_{PWR}) parameter space and look more closely at the time development of internal parameters of interest. First, however, we need to discuss the uncertainties due to three features built into the model: volume averaging, Maxwellian electrons, and one and the same back-attraction probability β for metal and process gas ions. The model only calculates average quantities in the ionization region. Recent observations of spoke structures in three different HiPIMS devices [29–31] demonstrate that, within a device, there can be large local deviations from such average values. The best diagnosed spokes [31] have a locally higher electron density, and contain more energetic electrons. The ionization probability for a sputtered atom is therefore higher inside these spokes than between them. The effect on the average degree of metal ionization $\langle n_{M^+} / (n_M + n_{M^+}) \rangle$ is, however, not obvious. The spokes are narrow in azimuth and rotate with speeds in the range from 4 km s^{-1} to 10 km s^{-1} , higher than the typical sputter wind speed. Bursts of sputtered atoms, as expected from the higher ion current density inside a spoke, could therefore either pass through it or become shifted in azimuth and miss it. A second source for uncertainty is the electron energy distribution function, EEDF, which in HiPIMS discharges is not Maxwellian as assumed in the

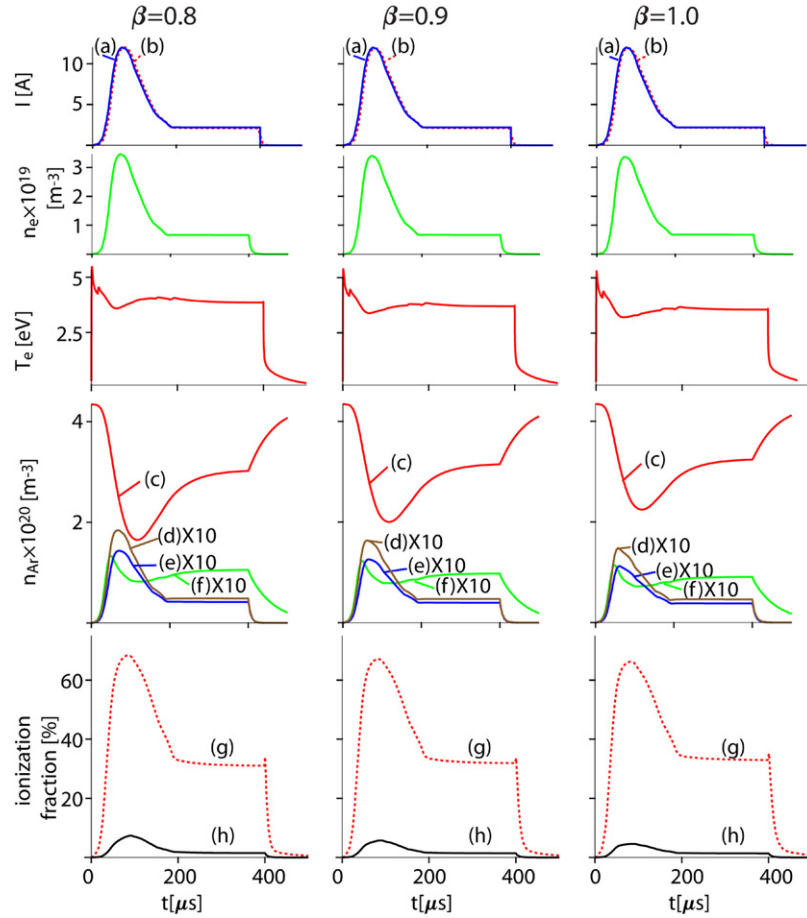


Figure 4. Overview of model results for different fitting parameters. In the three columns, from left to right, $\beta = 0.8, 0.9$ and 1.0 . For each β , F_{PWR} is chosen on the best fit curve (b)–(e) in figure 3. *First row:* (a) Experimental discharge current $I_D(t)$, and (b) modeled current $I_{\text{calc}}(t)$. *Second row:* $n_e(t)$. *Third row:* $T_e(t)$. *Fourth row:* (c) Total neutral argon gas density n_{Ar} . (d) Argon ion density n_{Ar^+} . (e) Hot recombined argon density n_{ArH} (in these runs, $n_{\text{ArW}} = 0$). (f) Metastable argon density n_{Ar^m} . *Fifth row:* (g) Degree of ionization of the sputtered species. (h) Degree of ionization of argon.

IRM. A high-energy tail is constantly replenished by energetic secondary electrons, and drained on a time scale $\tau_{\text{tail}} < \mu\text{s}$ mainly through ionization collisions. The likely effect of disregarding this tail in the IRM is an overestimate of the ratio between the metal and argon ionization rate coefficients $k_{i,M}/k_{i,\text{Ar}}$. Since the IRM is fitted through equation (1) to give the correct total ion flux to the target, the real degree of metal ionization $\langle n_{M^+}/(n_M + n_{M^+}) \rangle$ is therefore probably a bit lower than that obtained in the model. Finally, there is the assumption of one and the same β for process gas and sputtered species ions. In the discharge modeled here the known potential drop is $U_{\text{IR}} \approx 50$ V across the ionization region, which motivates high β values for both species, as discussed above. The assumption of a common β is therefore, in this case, probably only a minor source of error.

4. Evaluation of parameters during a pulse

Figure 4 gives an overview of the model calculations in the range $0.8 \leq \beta \leq 1$. The top row shows the obtained agreement between $I_{\text{calc}}(t)$ and $I_D(t)$, and the following rows show data that will be used in analyzing different aspects of the discharge.

4.1. Gas rarefaction

The development of the argon gas density during the pulse is shown in row 4 of figure 4. The overall trends are the same for all three β values modeled: the rate of gas rarefaction dn_{Ar}/dt is the fastest at the maximum in discharge current, and a highest absolute value of the gas rarefaction appears 40–60 μs after the discharge current maximum. Then, the gas density increases again to a constant value during the current plateau phase. After the pulse is turned off the gas refills and, with a time constant of 100–120 μs , returns to the initial value $n_{\text{Ar},0}$. Both the peak rarefaction and the steady value during the plateau phase vary with β in such a way that lower β corresponds to larger rarefaction. The significant rarefaction of Ar gas $\Delta n_{\text{Ar}}/n_{\text{Ar},0}$ during the HiPIMS discharge pulse is around 50%. This is consistent with working gas rarefaction which has been observed experimentally for a HiPIMS discharge [32, 33]. Using optical emission spectrometry Vlček *et al* [33] see almost an order of magnitude decrease in the density of atomic argon but a slightly less dramatic decrease in the density of argon ions at roughly 50–70 μs into a 200 μs long pulse while sputtering a copper target.

The physics behind these trends can be identified by analyzing the time plots of the terms in the argon density

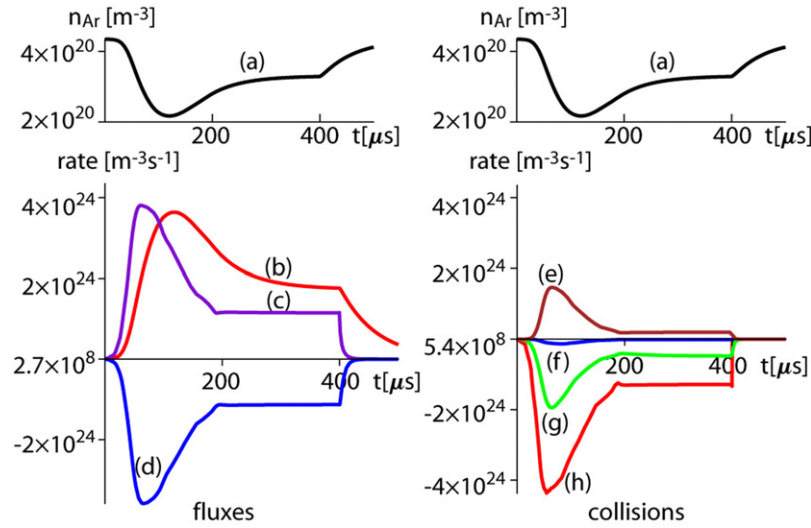


Figure 5. A study of the gas rarefaction and refill processes for the case $\beta = 0.9$. In both panels, curve (a) shows for reference the argon gas density from the fourth row in figure 4. In the left panel, the other curves show the fluxes across the borders of the ionization region expressed as a volume loss rate. Curve (b): diffusional refill of cold argon from the bulk plasma. Curve (c): returning Ar^{H} from the target. Curve (d): escaping Ar^{H} to the bulk plasma. The right panel shows the reaction rates for losses and gains of neutral argon atoms due to collisions inside the ionization region. Curve (e): $\text{Ar}^+ + M \rightarrow \text{Ar} + M^+$ charge exchange term $\times 10$. Curve (f): electron impact ionization of the hot Ar^{H} component. Curve (g): kick-out of cold argon atoms by the sputter wind. Curve (h): electron impact ionization of cold argon gas.

equations shown in figure 5. The cold gas density balance equation is in the IRM I given by [14]

$$\frac{dn_{\text{Ar}^{\text{C}}}}{dt} = -k_{\text{iz}}n_{\text{e}}n_{\text{Ar}^{\text{C}}} - k_{\text{exc},\text{m}}n_{\text{e}}n_{\text{Ar}^{\text{C}}} + k_{\text{chexc}}n_{\text{M}}n_{\text{Ar}^+} + k_{\text{p}}n_{\text{M}}n_{\text{Ar}^{\text{m}}} + k_{\text{dex}}n_{\text{e}}n_{\text{Ar}^{\text{m}}} + \frac{\Gamma_{\text{Ar,diff}}S_{\text{IR}}}{V_{\text{IR}}} - \frac{((\Gamma_{\text{M},0} - \Gamma_{\text{M,diff}}) + (\Gamma_{\text{M}^+,0} - \Gamma_{\text{M}^+,\text{diff}}))(S_{\text{IR}} - S_{\text{RT}})}{V_{\text{IR}}} - \frac{m_{\text{M}}}{m_{\text{Ar}}} \cdot \frac{n_{\text{Ar}^{\text{C}}}}{(n_{\text{Ar}^{\text{C}}} + n_{\text{Ar}^{\text{m}}})}, \quad (4)$$

where k_{nn} is the rate coefficient for process nn , and V_{IR} is the volume of the ionization region. For a detailed discussion of equation (4) the reader is referred to appendix III of Raadu *et al* [14]. The rather complicated last term represents kick-out by the sputter wind. Figure 5 shows the development of the terms in equation (4), and the corresponding equation for $n_{\text{Ar}^{\text{H}}}$, during a pulse with $\beta = 0.9$. The dominating loss terms are the electron impact ionization term $-k_{\text{iz}}n_{\text{e}}n_{\text{Ar}^{\text{C}}}$, curve (h), and the sputter wind kick-out, curve (g). Of these two, the ionization term is larger by more than a factor of two throughout the pulse. In order to illustrate why this is so, we consider the chain of events that begins when one argon atom becomes ionized and gets attracted to the target. We begin with an estimate of the sputter wind effect. With a sputter yield $Y_{\text{Ar}^+} \approx 1$, one sputtered atom is produced. For typical sputter energies of a few eV we take $\sigma_{\text{c}} = 2 \times 10^{-19} \text{ m}^2$ as a typical cross section for elastic neutral-neutral collisions [34]. Taking the degree of gas rarefaction from figure 4, the mean free path for the sputtered atom becomes $\lambda_{\text{c}} = 1/(\sigma_{\text{c}}n_{\text{Ar}}) \approx 1/(2 \times 10^{-19} \cdot 2 \times 10^{20}) \approx 0.025 \text{ m}$. The thickness of the ionization region is approximately 0.006 m. The probability for the sputtered atom to collide inside the ionization region is thus about $(1 - \exp(-0.006/0.025)) \approx 21\%$. Most of

the sputter wind would therefore pass through the ionization region without colliding with an argon gas atom. Let us compare this result with the direct ionization losses. All argon atoms that are ionized obviously disappear from the cold argon population. The fraction $(1 - \beta)$ that leaves to the bulk plasma is definitely lost, while the remaining fraction β hits the target, picks up an electron, and (for the moment assuming $n_{\text{Ar}^{\text{w}}} = 0$) returns to the ionization region as a part of the hot population $n_{\text{Ar}^{\text{H}}}$. This is seen as an influx, curve (c) in figure 5, which is smaller by about a factor β (here, 0.9) than the ionization losses, curve (h). They are approximated in the IRM [14] to enter the ionization region from the target with an equivalent temperature of 2 eV, giving an average speed of about 3 km s^{-1} . For the same reason as the sputtered species, the majority does not collide inside the ionization region and pass through it in about 2–4 μs . This gives a loss flux term, curve (d) in figure 5, which almost balances the influx in curve (b). The result is that argon atoms can be regarded as in practice lost upon ionization, as proposed earlier by Lundin *et al* [35]. It is also clear that the long mean free path is the basic reason why rarefaction by ionization losses dominates over the sputter wind in the discharge we model. For higher working gas pressures the sputter wind contribution would become more important.

Under what conditions, and when during a known HiPIMS pulse, can one expect severe gas rarefaction? To address these questions we introduce a hypothetical ionization loss rarefaction time $t_{\text{rarefaction}}$ from the pulse start, under conditions (1) that all produced argon ions go to the target, (2) that they carry the whole discharge current, (3) that the returning recombined argon atoms are quickly lost to the bulk plasma and (4) that there is negligible refill of gas. Complete gas rarefaction then would result when each of the argon atoms that were originally in the ionization region has transported

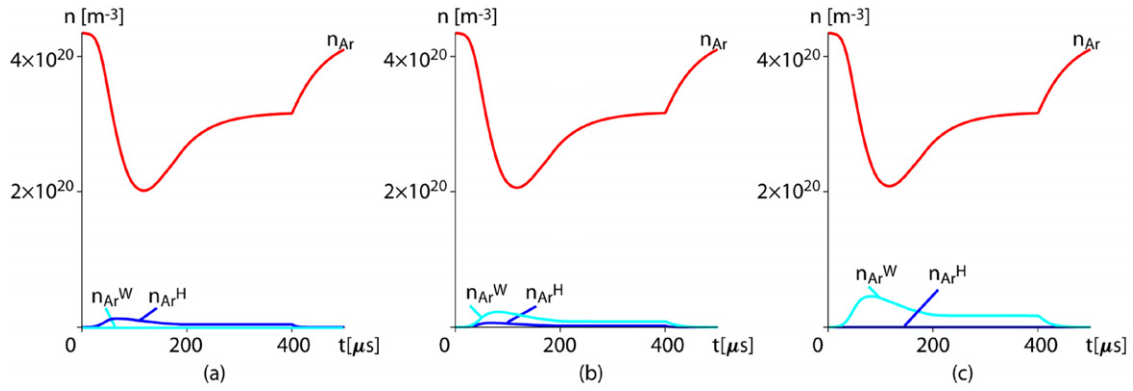


Figure 6. An investigation on the fate of the argon ions after hitting the target and recombination. Panel (a) is case 1, in which 100% of the argon atoms are reflected after recombination with typically 2 eV energy. Panel (b) represents case 2, where 50% are reflected and 50% penetrate. Case 3 is in panel (c), where 100% of the argon atoms penetrate.

the charge e to the target, and when this total charge equals the integrated discharge current, or

$$n_{Ar,0} V_{IR} e = \int_{t_{start}}^{t_{rarefaction}} I_D(t) dt. \quad (5)$$

In our case, $n_{Ar,0} \cdot V_{IR} \cdot e = 4.5 \times 10^{20} \text{ (m}^{-3}\text{)} \times 6.35 \times 10^{-6} \text{ (m}^3\text{)} \times e = 4.6 \times 10^{-4} \text{ (As)}$. Evaluating equation (5) with the experimental curve $I_D(t)$ gives a rarefaction time $t_{rarefaction} \approx 70 \mu\text{s}$, in fair agreement with the time $\approx 120 \mu\text{s}$ to maximum gas rarefaction in figure 4.

It is apparent from the results above that a key question for the gas rarefaction mechanism is the fate of the argon ions after hitting the target. We have up to now been assuming that all ions return immediately and add to the hot population n_{Ar^H} . There is also the possibility that they penetrate the target surface, displace atoms and come to rest below the surface. In the case of argon ions, such stopped ions are below the target surface without being bonded—a feature of the noble gas. Being weakly bound, they can diffuse to the surface and escape, adding to the warm population n_{Ar^w} . As a lower bound for the energy of the warm argon population we have taken it to be the thermal energy of the surface, of the order of 0.1 eV. However, the time of release from the target, in relation to the pulse-on time, must also be considered. If embedded, argon atoms diffuse out from the target at a constant rate, most of them would escape between pulses, and would only marginally counteract the gas rarefaction process during the pulse. The other extreme, escape preferentially during the pulse, would on the other hand probably correspond to higher effective n_{Ar^w} temperatures, since the bombardment of energetic ions somehow must be involved. We have tentatively investigated three scenarios: two extremes, and one intermediate. In case 1, 100% of the argon ions are reflected with typically 2 eV energy, i.e. the same case as in figures 4 and 5. In case 2, 50% are reflected and 50% penetrate, and in case 3, 100% penetrates. In cases 2 and 3 the escape of the penetrated argon atoms is assumed to be during pulse-on time, with an average energy of 0.1 eV, and at a rate proportional to the momentary current. Case 1 therefore gives a lower limit, and case 3 an upper limit to the density $n_{Ar^H} + n_{Ar^w}$. In all three cases the

model was run with $\beta = 0.9$. Excellent current fits (not shown) were obtained with the $F_{PWR} = 0.099, 0.100$ and 0.102 in cases 1, 2, and 3 respectively. Figure 6 shows the total argon density n_{Ar} , the warm recombined (penetrated) argon density n_{Ar^w} and the hot recombined (reflected) argon density n_{Ar^H} , for the three cases. Not even in case 3 does the recombined argon population become comparable to the cold argon density.

We can also note that the gas rarefaction process might take another route at lower pressures. The 1.8 Pa used here is unusually high for a HiPIMS discharge. If the experiments had been run with, say, half this argon density, but somehow forced to carry the same current $I_D(t)$, then the gas rarefaction time as estimated by equation (5) would be so short that a severe rarefaction of the cold argon gas component would be expected. The warm, recycled, component (see figure 6(c)) might then replace it as the dominating process gas component. This situation has been described by Anders *et al* [36] as a gas ‘recycling trap’ mechanism. More detailed understanding of the reflection, penetration, sputtering and diffusion processes of the recycled process gas ions is needed in our opinion to understand cases where the gas rarefaction time according to equation (5) is shorter than the high-current pulse duration.

Now let us turn to times after the peak current, during the current plateau phase. The data in figure 5 show that the degree of gas rarefaction is dominated here by the ionization losses, curve (h), and the diffusional refill term, curve (b). In the first approximation there is a particle balance here in which argon atoms diffuse into the ionization region from the surrounding volume, become ionized, and are attracted to the target where they recombine and return as hot atoms which pass quickly through the ionization region back to where they came from. The density difference between the surrounding volume and the ionization region (i.e. the degree of rarefaction) adjusts to such a value that the diffusional influx of neutrals is the same as the flux of argon ions to the target. After pulse-off, ionization stopped immediately and the refill process dominates. Therefore, the gas rarefaction is inversed and during a time constant of around $100 \mu\text{s}$, the argon density returns to $n_{Ar,0}$.

4.2. Electron density n_e , electron temperature T_e and the ionization fraction of the sputtered species

The temporal variation of the plasma density $n_e(t)$ is shown in the second row of figure 4. The electron density curves closely follow the discharge current in the top row. This is natural because, the IRM calculates the electron density in the ionization region only, within about 6 mm from the target surface. The average ion residence time in the ionization region can be extracted from the model as $t_{\text{res}} \approx n_e V_{\text{IR}} / (S_{\text{IR}} (\Gamma_{\text{Ar}^+} + \Gamma_{M^+}))$. It is typically a few μs . For this reason the plasma density can be approximated as the momentary ionization rate ($\text{m}^{-3} \text{s}^{-1}$) multiplied by t_{res} . Since the energy input, and therefore the ionization rate, approximately follows the discharge current, the momentary plasma density in the ionization region also follows the discharge current. As a consequence, the absolute densities are almost independent of the variations in the fitting parameters β and F_{PWR} . All three curves show peak plasma densities of $n_e = 3.5 \times 10^{19} \text{m}^{-3}$, which coincide with the current maxima, and plateaux at $n_e = 6.5 \times 10^{18} \text{m}^{-3}$. This is in line with experimental results on HiPIMS discharges, which show peak plasma densities exceeding 10^{19}m^{-3} in the close vicinity of the target [37]. In the adjacent bulk plasma and in the anode region slower time variations of the plasma density are expected, as also seen by measurements [4, 37, 38]. These regions are mainly supplied by, and accumulate, plasma flowing from the ionization region.

The electron temperature $T_e(t)$ (the third row in figure 4) is always around 4 eV during the plateau, and with a small variation during the high-current transient. Also regarding $T_e(t)$ it must be pointed out that this is a model of the ionization region only. There is experimental evidence [4, 21] of much lower electron temperatures in the bulk plasma and in the plasma in the anode region. The natural explanation is that in these regions there is no direct inflow along the magnetic field lines B of energetic secondary electrons from the target, and thus no or very limited ionization taking place. Note also that, after pulse-off, the electron temperature drops below 1 eV in less than 10 μs . Plasma drifting from the ionization region into the bulk plasma and cut-off from the energy source would cool off at a similar rate. This agrees with optical measurements by Hala *et al* [21], which show a separation between an ionization region with intense light emission, and a much darker surrounding bulk plasma.

The ionization fraction $\langle n_{M^+} / (n_M + n_{M^+}) \rangle$ of the sputtered species is shown in the bottom row of figure 4. It varies only a little with β and F_{PWR} , and therefore can be taken as one of the more reliable results of the model. It peaks around 70% during the high-current peak, and then stabilizes a little above 30% during the plateau phase. An inspection of the involved terms shows that this time variation is mainly due to the higher electron density during the peak, which gives a higher probability of ionization during the time an average sputtered atom stays in the ionization region. It can be noted that the metal ion fraction is about an order of magnitude larger than the argon ion fraction, as seen in figure 4, bottom panel. The main reason is the difference in the ionization potential, 5.96 eV for aluminum and 15.7 eV for argon.

The threshold for producing doubly ionized aluminum through the reaction $e + M^+ \rightarrow 2e + M^{2+}$ is 18.7 eV, even higher than the 15.7 eV required for the ionization of argon. From the cross section for this reaction [39, 40] we have made some preliminary runs of the IRM including also M^{2+} , giving the estimated fraction to be at most 2%, justifying that it is not included in these calculations.

4.3. Degree of self-sputtering

In Anders' experiments [36], the results were analyzed in terms of the self-sputtering parameter [36, 41, 42],

$$\tilde{\Pi} \equiv \sum_s \tilde{\alpha}_s \tilde{\beta}_s \tilde{\gamma}_s, \quad (6)$$

which describes the particle balance for the sputtered species, as illustrated in figure 7(a). In their analysis the $\sim$ sign indicates the generalized character of the neutral fluxes from the target, which contained both gas and metal species in most cases. $\tilde{\alpha}_s$ is the effective probability of ionization of the generalized sputtered species, $\tilde{\beta}_s$ is the effective probability that the newly formed ion returns to the target and $\tilde{\gamma}_s$ is the generalized self-sputter yield of the ions. Self-sustained self-sputtering requires $\tilde{\Pi} = 1$, whereas the discharge goes into self-sputtering runaway when $\tilde{\Pi} > 1$. Since both $\tilde{\alpha}_s$ and $\tilde{\beta}_s$ have upper limits of 1, self-sustained self-sputtering requires $\tilde{\gamma}_s > 1$, with a margin to compensate for $\tilde{\alpha}_s$ and $\tilde{\beta}_s$ below unity. This is most easily achieved in target materials with very high sputter yields such as Cu and Ag. For other materials, self-sustained self-sputtering can still occur [13] if the applied power is high enough to produce twice ionized metal M^{2+} . These ions are necessary for two reasons; first, they emit the secondary electrons necessary to sustain the discharge (which M^+ ions do not in this energy range) and, second, they are accelerated toward the target to twice the energy, which boosts their self-sputter yield.

To apply the self-sputtering parameter $\tilde{\Pi}$ to our model result, we approximate the probability of metal ionization α by the degree of ionization calculated in the model, i.e. $\alpha \approx n_{M^+} / (n_M + n_{M^+})$. γ is known as a function of the ion bombarding energy, which we approximate as $E_i \approx eU_D$ [43], and β is a model input parameter. Figure 7(b) shows the development during a pulse, demonstrating that $\Pi < 1$, always well below the threshold for self-sustained self-sputtering as expected. Figure 7(c) shows the fraction of the ion current to the target that is carried by metal ions, $I_{M^+} / (I_{\text{Ar}^+} + I_{M^+})$. As can be seen, it nicely follows the development of the self-sputtering parameter in figure 7(b).

Figure 7(d) illustrates an increasing degree of self-sputtering with increasing discharge power, as proposed and qualitatively explained by Anders *et al* [13]. Here, the model has been run for all the discharge current curves measured for target voltage below 450 V shown in figure 1, including other even lower voltages that are not plotted in the figure. In all cases the back-attraction probability has been set to $\beta = 0.9$. The metal ion current fraction $I_{M^+} / (I_{\text{Ar}^+} + I_{M^+})$, which is shown in figure 7(d), has then been extracted in the middle of the plateau phase. The curve in figure 7(d) is limited to $U_D \leq 450$ V since, for higher discharge powers, we regard it to be too uncertain because doubly ionized metal is not included in the model.

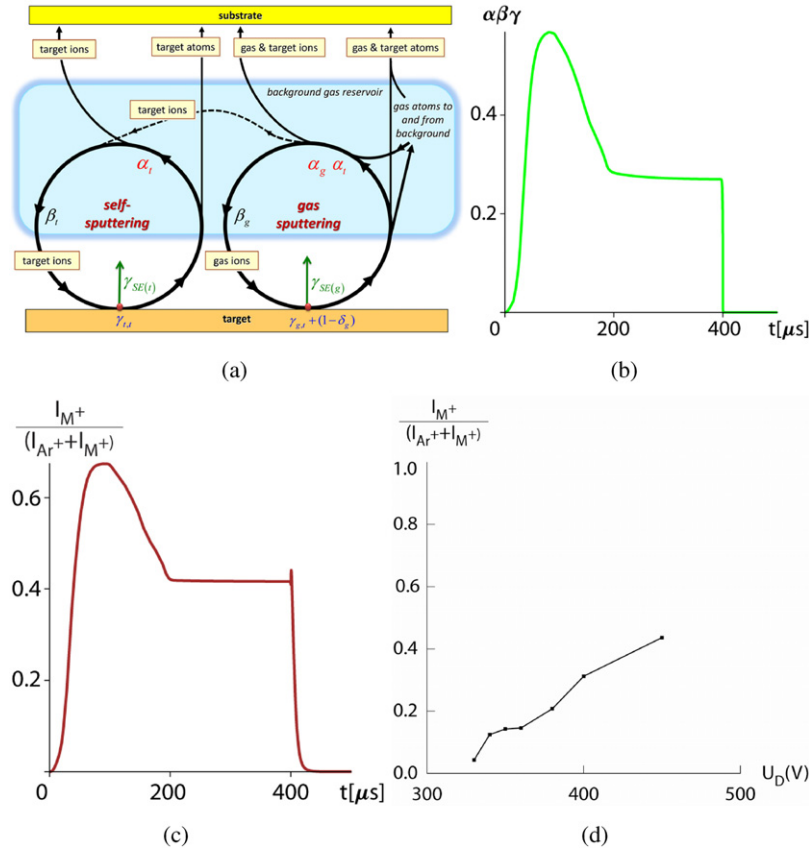


Figure 7. (a) Self-sputter feedback loop (from [36]). (b) Time development of the self-sputtering parameter $\tilde{\Pi} \equiv \sum_s \tilde{\alpha}_s \tilde{\beta}_s \tilde{\gamma}_s$ during a pulse with $U_D = 450 \text{ V}$ and $\beta = 0.9$. (c) Metal ion fraction of the ion current to the target, $I_{M^+}/(I_{Ar^+} + I_{M^+})$, during the same pulse. (d) Fraction $I_{M^+}/(I_{Ar^+} + I_{M^+})$ in the plateau phase, $300 \mu\text{s}$ into the pulse, as a function of the discharge voltage. Also here, $\beta = 0.9$.

4.4. Deposition rate variation during the pulse

Several mechanisms have been proposed to contribute to give a lower deposition rate in HiPIMS compared with dcMS at the same average power [1]. Some of the mechanisms are based on well-known effects such as a higher energy cost per sputtered particle for the higher HiPIMS ion bombarding energies [44], a close to zero secondary electron emission probability for singly charged metal ions [13], and, for some species, a low self-sputtering yield [1, 45]. Less well understood mechanisms involve the large gas rarefaction in HiPIMS which reduces the number of ions available for sputtering, and the electric back-attraction of ions to the target [15, 46, 47].

Here we will limit the discussion to the last effect, in the model represented by the β value. We quantify this effect using the model as follows. In the long run, all sputtered particles $(n_M + n_{M^+})V_{IR}$, which at a given moment are in the ionization region, leave it. Of these, a fraction βn_{M^+} of the ionized component M^+ goes to the target. The neutral sputtered atoms M , on the other hand, all start with a velocity directed away from the target. As calculated in section 4.1, with a typical degree of gas rarefaction (see figure 4), their mean free path in the 450 V pulse is 25 mm. Only 21% of those atoms would collide in the 6 mm thick ionization region and, if they after collision leave it isotropically, typically one-third (or 7% of all neutrals) would return to the target. We neglect this small fraction. The reduction in the flow of sputtered species

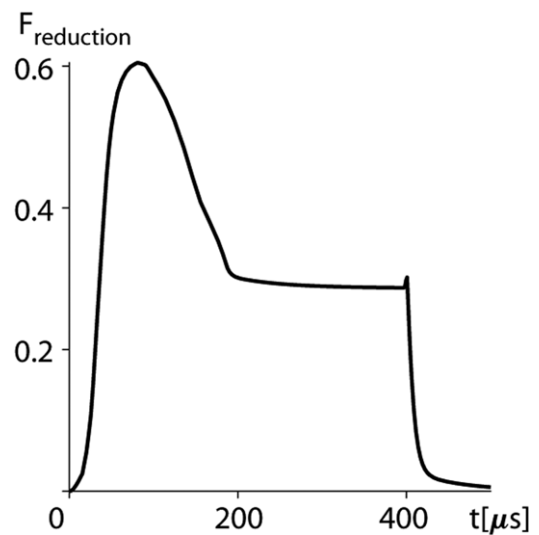


Figure 8. An estimate of the part of the reduction of the deposition rate that can be attributed to electric back-attraction of ions.

due to back-attraction of the charged component can then be approximated by a factor

$$F_{\text{reduction}} \approx \frac{\beta n_{M^+}}{n_M + n_{M^+}}. \quad (7)$$

This factor is extracted from the standard model run ($\beta = 0.9$, $n_{Ar^w} = 0$) and shown in figure 8. It first shows a rising trend

from an initial low value. This is consistent with experimental results of Konstantinidis *et al* [48] who found that in the case of Ti sputtering, approximately 70% of the dcMS deposition rate was obtained for 5 μ s long pulses, compared with 20% of the dcMS rate for 20 μ s long pulses. Although the trend seen in figure 8 is consistent with this development at the beginning of the pulse, it also shows that the discharge might return to higher deposition rates for long enough pulses. Furthermore, deposition rate studies by Samuelsson *et al* [49] sputtering Al in an Ar atmosphere using 100 μ s HiPIMS discharge pulses show that 49% of the dcMS rate is achieved, in line with the results shown here. The pulse lengths often used in the early HiPIMS devices, typically 50–150 μ s [1, 2, 4], might actually be close to the worst choice with respect to the deposition rate. A deeper understanding of the development during a pulse opens a possibility to optimize the deposition rate by choosing an appropriate pulse length. Complications that need to be considered are that β is likely to be different between Ar⁺ and ionized sputtered particles M^+ due to the directionality of M^+ , as well as the very likely possibility that β varies in time.

5. Discussion and summary

From the model studies made of long discharge pulses in HiPIMS using the ionization region model (IRM) it can be concluded that the plasma chemistry as well as the plasma characteristics can be tracked in time in the ionization volume located above the target race track. The IRM was adjusted to reproduce a HiPIMS experiment performed by Anders *et al* [13] sputtering Al in an Ar atmosphere by fitting the geometry and using a reaction set for the working gas and target species. In addition, the model uses two fitting parameters, the probability β of ion back-attraction to the target, and the fraction F_{PWR} of the electric discharge power, $P_{\text{D}} = U_{\text{D}}I_{\text{D}}$, that goes to the electrons. Using a best fit scheme, model calculations were found to give a close fit to the experimentally observed initial high-current transient, as well as the later plateau value of constant lower current, constricting the possible parameter ranges to $0.5 < \beta < 1$ and $0.1 < F_{\text{PWR}} < 0.156$. These ranges are in agreement with independent estimates, $0.8 < \beta < 1$ and $0.075 < F_{\text{PWR}} < 0.112$, obtained from the experimentally determined potential profiles. The uncertainty in the latter estimate of F_{PWR} mainly comes from the uncertainty in the recapture probability r of secondary electrons. In the identified range of fitting parameters, the mechanisms affecting the ionization of Ar as well as sputtered Al, the deposition rate and gas rarefaction were studied. It was found that during the high-current transient of the HiPIMS discharge ($t \sim 100 \mu$ s), the degree of sputtered metal ionization reaches 65–75%, and then drops to 30–35% during the current plateau phase ($200 < t < 400 \mu$ s). This time evolution is shown to be mainly related to the evolution of $n_e(t)$, which in turn was seen to be directly correlated with the discharge current. The loss in deposition rate that can be attributed to the back-attraction of the ionized sputtered species was also estimated from the model. It is low during the initial 10–20 μ s, peaks around 60% during the high-current transient, and finally stabilizes

around 30% during the plateau phase. Furthermore, it was found that there is a significant rarefaction of Ar gas during the HiPIMS discharge pulse ($\Delta n_{\text{Ar}}/n_{\text{Ar},0} \approx 50\%$). From the model it is also found that the dominating mechanism for gas rarefaction is ionization losses, with only about 30% due to the sputter wind kick-out process. For higher working gas pressures the sputter wind contribution would become more important. Finally, the degree of self-sputtering (defined here as the metal ion fraction of the total ion current to the target) was seen to vary during the pulse with a peak of approximately 70–75% during the high-current transient. For future work it would be very interesting to advance into the runaway self-sputtering dominated regime, characterized by a second current increase after the initial current peak. This implies introducing multiply charged sputtered material ions into the IRM code, since they possess high enough potential energy to release enough secondary electrons to sustain a highly ionized plasma.

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